

Automated Pneumothorax Detection and Segmentation from Chest X-rays Using Multi-task Learning

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Abstract—Pneumothorax is a potentially life-threatening condition characterized by air accumulation in the pleural cavity, leading to collapsed lungs. Early and accurate diagnosis through chest X-rays is critical yet often complicated by human error and diagnostic variability. Automating pneumothorax detection and segmentation using artificial intelligence (AI) techniques can significantly reduce diagnostic delays and inaccuracies, thus facilitating quicker and more effective patient care. In this work we propose a novel U-net-based architecture for segmentation and detection of pneumothorax from X-ray images. We have used the Efficient Net backbone encoder and Unet decoder for segmentation, with intermediate encoder and decoder feature outputs feeding to the classifier instead of conventionally using the final encoder output.

Index Terms—U-net, image segmentation, deep learning, medical imaging, clinical decision support, Efficient-Net

I. INTRODUCTION

Motivation: Pneumothorax is a potentially life-threatening condition caused by the accumulation of air in the pleural cavity, resulting in the partial or complete collapse of the lung. This collapse compromises oxygen exchange, leading to hypoxia and, in severe cases, can escalate into tension pneumothorax—a critical state in which air continues to accumulate, compressing the heart and major vessels, and possibly culminating in circulatory shock and cardiac arrest [1]. The condition is particularly prevalent in trauma care, affecting over 20% of patients with major blunt injuries [2]. Prompt and accurate detection is therefore essential, particularly for mechanically ventilated or polytrauma patients, where diagnostic delays can rapidly exacerbate respiratory and hemodynamic instability.

Although computed tomography (CT) and ultrasound offer high sensitivity, chest radiography remains the most accessible and cost-effective imaging modality, especially for routine

post-interventional monitoring. However, its diagnostic accuracy is hindered by inter-observer variability and low sensitivity in early-stage pneumothorax cases. Recent studies have demonstrated the potential of deep learning (DL) to overcome these limitations. For example, a DL model trained on large-scale datasets generalized effectively across diverse external cohorts, highlighting its robustness across demographic and technical variations [3]. Hong et al. [4] further showed that their DL model outperformed radiologists in detecting pneumothorax post-percutaneous biopsy, notably improving sensitivity for small-volume pneumothoraces (less than 15%) [3]. These results underscore the promise of AI in enhancing early detection and mitigating diagnostic errors in clinical workflows.

Problem statement: Despite recent advancements, integrating AI tools for real-time, accurate pneumothorax detection and segmentation remains a challenge in clinical settings. Rapid and precise localization of pneumothorax on chest radiographs is critical for initiating timely interventions, yet existing approaches often require extensive labeled data or fail to generalize across heterogeneous patient populations.

To address this, several prior works have explored weakly supervised and multi-task learning frameworks. For instance, Rajpurkar et al. [5] proposed a deep neural network capable of multi-label classification and pathology localization from frontal chest X-rays. Building on this, Liu et al. [6] developed a two-module architecture incorporating an encoder-decoder network for localizing the dehiscence visceral pleura and a DenseNet-inspired classifier [7] for pneumothorax prediction. Abedalla et al. [8] have introduced segmentation approaches with ResNet and Efficient-Net encoders for isolating the affected pneumothorax regions. While these models demonstrated promise, there remains a need for unified architectures

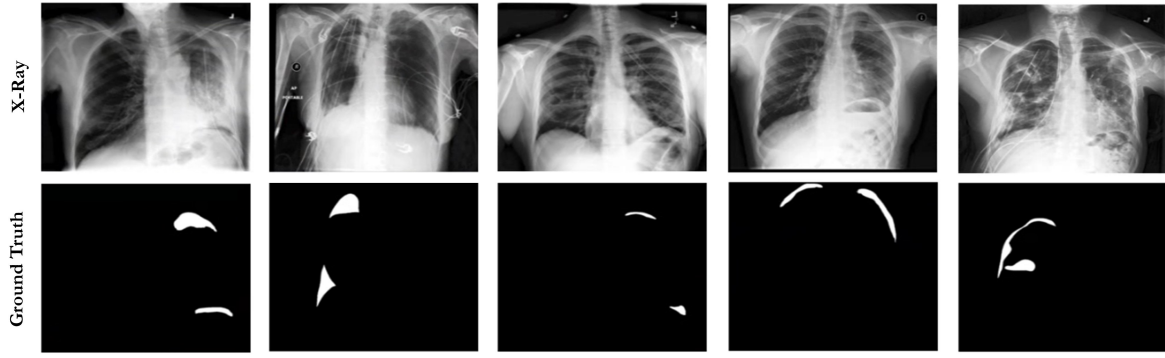


Fig. 1. Some example cases of pneumothorax across the dataset.

that integrate both segmentation and classification to enhance diagnostic accuracy and reduce clinical burden.

In this project, we propose a multi-task learning framework that jointly performs segmentation and classification for pneumothorax detection. By integrating an Efficient-Net backbone as a feature-extracting encoder and a conventional U-Net-based decoder, the segmentation pipeline is formulated. The final few layers of the concatenated encoder and decoder features are globally averaged and passed through fully connected neural networks to design the classification pipeline. By training both pipelines in unison with a multi-task loss function, we obtain pneumothorax predictions that can aid in clinical applications.

II. DATASET AND EXPERIMENTAL SETUP

A. Data

The dataset used in this project is sourced from the SIIM-ACR Pneumothorax Segmentation competition organized by the Society for Imaging Informatics in Medicine (SIIM). It provides chest X-ray images with corresponding segmentation masks and binary class labels indicating the presence or absence of pneumothorax. The dataset is divided into Train and Test sets, as shown in Table I. To build the model, the Train set was further split into 80% training and 20% validation subsets. To address class imbalance, since ‘No Pneumothorax’ cases significantly outnumber ‘Pneumothorax’ cases, we performed sub-sampling to balance the training data. The final balanced distribution for training and validation is shown in Table II. Furthermore, some example pneumothorax cases have been selected and are depicted in Fig. 1.

TABLE I
DISTRIBUTIONS OF THE SIIM-ACR DATA

Category	Train	Test
Pneumothorax	2379	290
No Pneumothorax	8296	1082

The utilized dataset link has been provided here for download: <https://www.kaggle.com/datasets/vbookshelf/pneumothorax-chest-xray-images-and-masks>

TABLE II
TRAIN AND VALIDATION DISTRIBUTION AFTER BALANCING

Category	Train	Val
Pneumothorax	944	1435
No Pneumothorax	950	700

B. Setup

To ensure consistency in model training, all images and masks were first resized to a fixed resolution of 256×256 pixels. This resizing step standardizes the input dimensions, making it suitable for batch processing and ensuring compatibility with the network architecture. In addition to resizing, histogram equalization was applied to the grayscale chest X-ray images. This preprocessing step helps improve contrast by redistributing pixel intensity values to achieve a more uniform histogram. This makes small or low-contrast pneumothorax regions more visible to the model. During training, data augmentation techniques were applied to improve model generalization and prevent overfitting. These augmentations included random rotations of up to $\pm 10^\circ$ and random translations of up to $\pm 10\%$ of the image dimensions in both horizontal and vertical directions.

III. METHODS

A. Architecture

We propose a multi-task learning approach that combines segmentation and classification using a modified U-Net architecture with EfficientNet backbones [9]–[11]. Our model consists of two main components: (1) an encoder-decoder network for pneumothorax segmentation and (2) a classification branch that leverages intermediate encoder features for pneumothorax detection.

For the encoder module, we experimented with different backbone architectures, including ResNet34, EfficientNet-B4 and B5. The EfficientNet family of models was chosen for its state-of-the-art performance compared to other convolutional neural networks [10] and for its ability to effectively identify and extract useful feature context. For our case, the encoder extracts hierarchical features from the input chest X-rays,

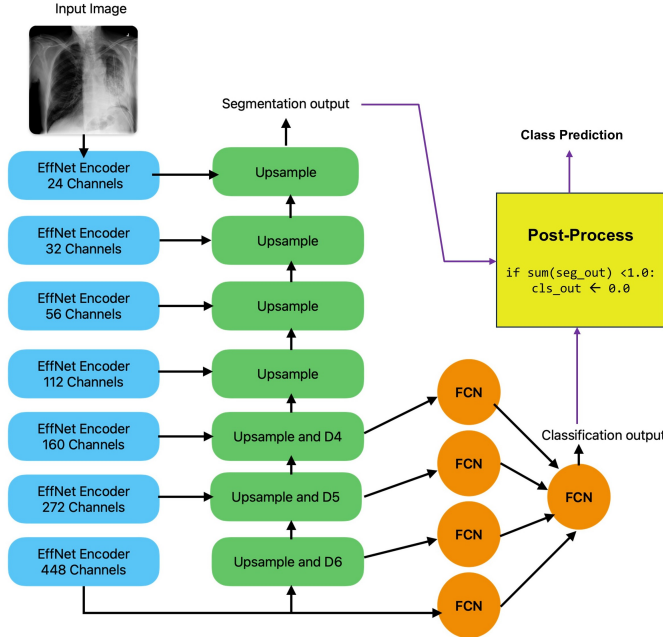


Fig. 2. Architecture of our project to perform both segmentation and classification.

capturing both fine-grained details and high-level semantic information.

The decoder module follows the traditional U-Net architecture [9] with skip connections that combine features from the encoder at different resolutions. This helps preserve spatial information that is critical for accurate segmentation of pneumothorax regions, which can vary significantly in size and location. The decoder progressively upsamples the feature maps to generate a full-resolution segmentation mask.

An important step in our approach is the classification branch that takes intermediate feature maps from the encoder as well as decoder, and processes them through fully connected layers (FCN) to predict the presence or absence of pneumothorax in an input image. These branches share parameters with the segmentation pathway during training, allowing the model to learn representations that are beneficial for both tasks simultaneously. This multi-task approach is similar to that proposed by Kendall et al. [11].

For an input chest X-ray image x , our model produces:

- A segmentation output $S(x) \in [0, 1]^{H \times W}$, where H and W are the height and width of the input image
- A classification output $C(x) \in [0, 1]$ indicating the probability of pneumothorax presence

B. Loss Function

To optimize both segmentation and classification tasks simultaneously, we developed a composite loss function that combines binary cross-entropy (BCE) for classification and Jaccard loss [12] for segmentation:

$$\mathcal{L}_{total} = \mathcal{L}_{Jaccard}(S(x), y_{mask}) + \gamma \cdot \mathcal{L}_{BCE}(C(x), y_{class}) \quad (1)$$

where y_{class} is the binary label indicating pneumothorax presence, y_{mask} is the ground truth segmentation mask, and γ is a weighting factor that balances the contribution of the segmentation loss. We empirically set $\gamma = 0.01$ to ensure that the classification task does not dominate the learning process, following principles similar to those described by Kendall et al. [11] for multi-task learning.

The Jaccard loss, based on the Jaccard similarity coefficient (also known as Intersection over Union or IoU), is common in medical imaging. The Jaccard loss is defined as:

$$\mathcal{L}_{Jaccard} = 1 - \frac{\sum_i p_i g_i}{\sum_i p_i^2 + \sum_i g_i^2 - \sum_i p_i g_i} \quad (2)$$

where p_i is the predicted probability for pixel i and g_i is the corresponding ground truth label. This formulation directly optimizes the IoU metric, which is more appropriate for our specific task of pneumothorax segmentation than other loss functions.

C. Post-Processing

Post-processing techniques were applied to refine the model's segmentation outputs. After obtaining the raw segmentation predictions, we applied a threshold of 0.5 to binarize the probability maps.

A critical component of our post-processing pipeline is a conditional logic rule that helps reduce false positive classifications. As shown in Figure 2, if the sum of all predicted pixel values in the segmentation output is below a threshold of 1.0 (representing a very small or negligible pneumothorax area), the classification output is automatically set to zero:

$$\text{If } \sum_{\text{all pixels}} S(x) < 1.0, \text{ then } C(x) = 0.0 \quad (3)$$

This rule effectively prevents the model from classifying an image as containing pneumothorax when the segmented region is extremely small, likely representing noise rather than actual pathology. This post-processing step is particularly important in clinical settings where false positives could lead to unnecessary interventions [4].

IV. RESULTS AND DISCUSSION

We evaluate performance using both segmentation and classification metrics. Segmentation quality is assessed with Jaccard Index (IoU) and Dice scores. And, classification performance is measured using Accuracy, Precision, Recall, and F1-Score.

A. Qualitative Result

In the qualitative results shown in Fig. 3, we observe that the first four predictions (from left to right) achieve noticeably higher IoU scores (ranging from 0.74 to 0.85) compared to the latter two cases. This aspect is partially due to the distribution of positive pixels (i.e., foreground mask regions) in the ground truth annotations, as depicted in the histogram in Fig. 3. The histogram reveals a strong skew toward sparse annotations, with the majority of cases containing less than

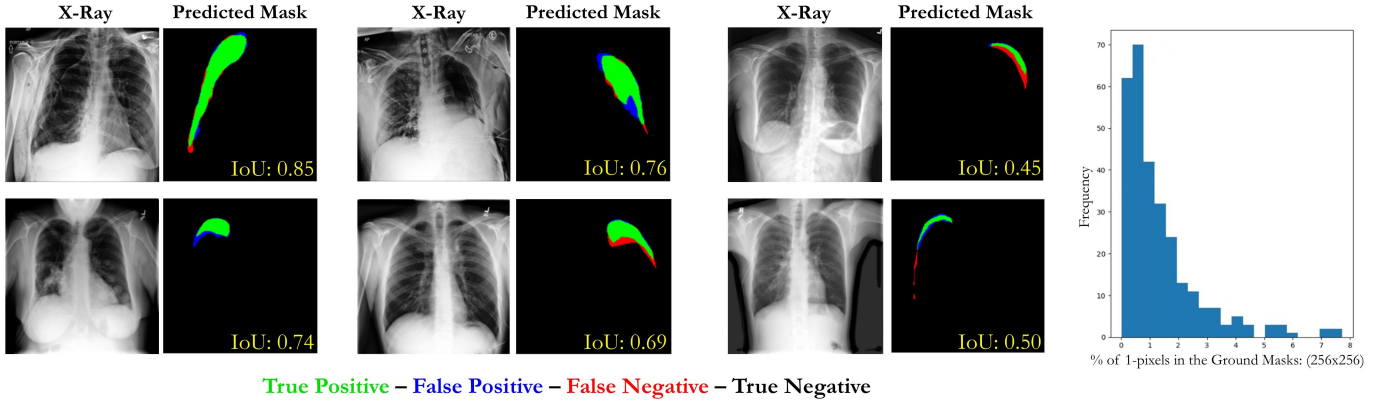


Fig. 3. Prediction results from segmentation and pixel distribution in the pneumothorax existing test set.

TABLE III
QUANTITATIVE RESULTS

Model Type	Classification Layers	Segmentation Metrics		Classification Metrics			
		Jaccard	Dice	Accuracy	Precision	Recall	F1-Score
Backbone: Res34Net, Decoder: Unet	No	0.27	0.29	-	-	-	-
Backbone: EfficientB4Net, Decoder: Unet	No	0.56	0.59	-	-	-	-
Backbone: EfficientB5Net, Decoder: Unet	No	0.59	0.61	-	-	-	-
Backbone: EfficientB4Net, Decoder: Unet	Yes	0.62	0.64	0.79	0.54	0.74	0.63
Backbone: EfficientB5Net, Decoder: Unet	Yes	0.69	0.71	0.81	0.62	0.78	0.69

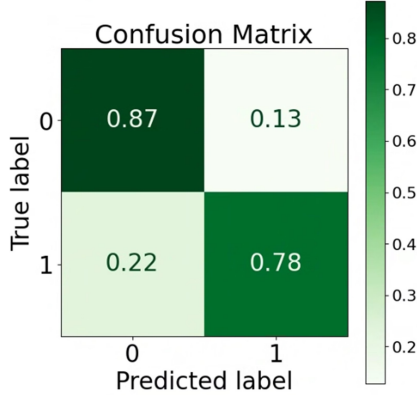


Fig. 4. Confusion Matrix of the proposed approach.

2% positive pixels out of the total 256×256 resolution. Such sparsity increases the likelihood of segmentation failure due to high level of performance degradation with just a few pixel mispredictions. On the other hand, examples with more extensive affected areas, such as left-hand four xrays from Fig. 3, depict clearer structural patterns and allows our model to achieve higher IoU. This observation underscores the difficulty of accurately delineating minimal pneumothorax regions, where even slight boundary errors can drastically degrade overlap-based metrics.

B. Quantitative Result

Table III summarizes the progression of model performance from the weakest to the strongest configuration. The model

using Res34Net as backbone and Unet decoder without classification layers performs the worst. It achieved a Jaccard of only 0.27 and Dice of 0.29, indicating poor spatial agreement with the ground truth. Substituting Res34Net with EfficientNet backbones significantly improves performance even without classification heads—e.g., EfficientB4Net and B5Net achieve Jaccard scores of 0.56 and 0.59, respectively. This shows the value of stronger feature extractors.

The inclusion of classification layers further boosts performance across all metrics. The best configuration, EfficientB5Net with classification layers, achieves a Jaccard of 0.69, Dice of 0.71, and the highest classification accuracy of 0.81. This model also yields a precision of 0.62, recall of 0.78, and F1-score of 0.69. The corresponding confusion matrix in Fig. 4 shows these improvements, with high true negative (0.87) and true positive (0.78) rates. EfficientB5 backbone model reliably distinguishes between pneumothorax and normal cases. As a finding in our project, we have found that deeper backbones coupled with classification-guided supervision can lead to more accurate and generalizable predictions.

V. CONCLUSIONS

This work demonstrates a multi-task learning framework for automated pneumothorax detection and segmentation from chest X-rays. By leveraging EfficientNet-based encoders and a classification-guided architecture, the model achieves strong performance in the evaluation metrics. The inclusion of post-processing logic further enhances clinical reliability by reducing false positives. Despite promising results, future improvements could include exploring more advanced encoder-

decoder architectures like ConvNeXt or vision transformers to better capture spatial-semantic features. Additionally, incorporating patch-based training strategies may enhance model sensitivity to small or localized pneumothorax regions, especially in low-pixel-density cases. These directions hold potential to further improve performance and generalizability of the scope of this project.

VI. CONTRIBUTIONS IN WRITING

Mentioned below are the contributions in writing this project report among the four authors:

1st Author - Abstract, Introduction.

2nd Author - Methods.

3rd Author - Figures, Results and Discussion, Conclusion.

4th Author - Dataset and Experimental Setup.

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